

Signal and Systems

- Signal are represented mathematically as functions of one or more independent variables.
- Digital signal processing deals with the transformation of signal that are discrete in both amplitude and time.
- Discrete time signal are represented mathematically as sequence of numbers.

Signals and Systems

- A discrete time system is defined mathematically as a transformation or operator.

□ $y[n] = T\{x[n]\}$



Linear Systems

- The class of *linear systems* is defined by the principle of superposition.

$$T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_2[n]\} = y_1[n] + y_2[n]$$

- And

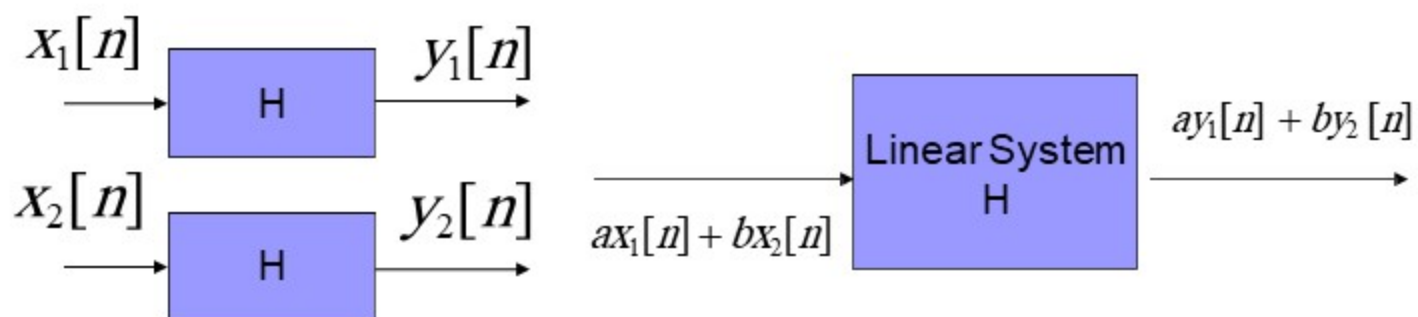
$$T\{ax[n]\} = aT\{x[n]\} = ay[n]$$

- Where a is the arbitrary constant.
- The first property is called the *additivity property* and the second is called the *homogeneity or scaling property*.

Linear Systems

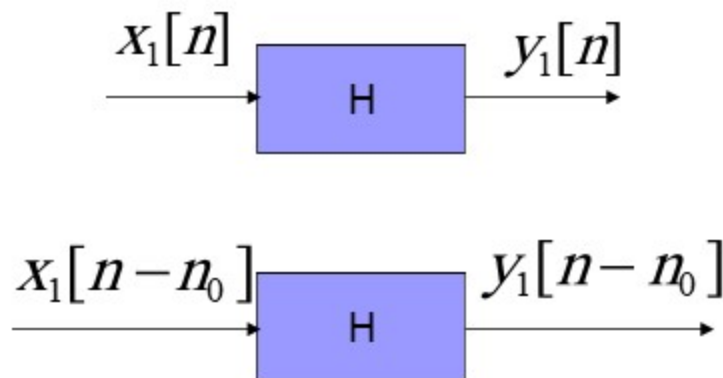
- These two property can be combined into the principle of superposition,

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$$



Time-Invariant Systems

- A Time-Invariant system is a system for which a time shift or delay of the input sequence cause a corresponding shift in the output sequence.



LTI Systems

- A particular important class of systems consists of those that are linear and time invariant.
- LTI systems can be completely characterized by their impulse response.

$$y[n] = T\left\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\right\}$$

- From principle of superposition:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]T\{\delta[n-k]\}$$

- Property of TI:

$$y[n] = \sum_{k=-\infty}^{K=+\infty} x[k]h[n-k]$$

LTI Systems (Convolution):

$$y[n] = \sum_{k=-\infty}^{k=+\infty} x[k]h[n-k]$$

- Above equation commonly called *convolution sum* and represented by the notation

$$y[n] = x[n] * h[n]$$